

create_keepout_margin

NAME

create_keepout_margin
Creates design and cell keepout margins.

SYNTAX

```
collection create_keepout_margin
  [-type hard | soft | hard_macro | routing_blockage | clock |
  seal_ring | assembly_die]
  [-outer margin_spec]
  [-inner margin_spec]
  [-layers layers]
  [-tracks_per_macro_pin number_of_tracks]
  [-min_padding_per_macro padding_size]
  [-max_padding_per_macro padding_size]
  [design_and_cell_list]
```

Data Types

margin_spec	list
layers	list
number_of_tracks	float
padding_size	float
design_and_cell_list	list

ARGUMENTS

- type hard | soft | hard_macro | routing_blockage | clock | seal_ring | assembly_die | filler_allowed
Specifies the type of the keepout margin to create. Valid values are
- o hard (the default)
A hard keepout margin prevents objects from being placed in the keepout margin area throughout the entire flow. However, it would not prevent hard macros being placed in the keepout margin area.
 - o soft
A soft keepout margin prevents the coarse placer from placing objects in the keepout margin area, but optimization and legalization can place objects within the keepout margin. Also, it would not prevent hard macros being placed in the keepout margin area.
 - o hard_macro
A hard macro keepout margin prevents hard macros from being placed in the keepout margin area throughout the entire flow.
 - o routing_blockage
A routing blockage keepout margin prevents PG routes from being placed in the keepout margin area.
 - o clock
A clock keepout margin prevents clocks from being placed in the keepout margin area.
 - o seal_ring
A seal_ring keepout margin around the block to prevent any shapes from being placed in the keepout margin area. Only outer keepout can be set for this type. The margin must be pointing outward (positive), and can only be placed on a block.
 - o assembly_die
An assembly_die keepout margin around the block or seal_ring keepout to prevents any shapes from being placed in the keepout margin area. Only outer keepout can be set. The margin must be pointing outward (positive), and can only be placed on a block. An assembly die keepout is relative to the seal_ring keepout if a seal_ring keepout is set, otherwise it is relative to the block.
 - o filler_allowed
A filler_allowed keepout margin is a hard keepout and prevents objects from being placed in the keepout margin area throughout the entire flow. However, it would not prevent filler cells or hard macros being placed in the keepout margin area.

-outer margin_spec
Specifies the padding margin to maintain outside of the design or cell boundary. This setting is used for physical designs and cells.

Use the following syntax to specify the margins:

{left bottom right top}

The margin parameters must be zero or positive and all four parameters are required. The directions are relative to the core area. The units are microns.

-inner margin_spec
Specifies the padding margin to maintain inside the design boundary. This option is not valid for leaf cells.

Use the following syntax to specify the margins:

{left bottom right top}

The margin parameters must be zero or positive and all four parameters are required. The directions are relative to the core area. The units are microns.

-layers layers
Specifies the layers for the routing_blockage type keepout. This option must be specified with, and can only be used with the -type routing_blockage keepout. Providing an empty layer list causes an error. Specify the layers by using the layer names from the technology file.

-tracks_per_macro_pin number_of_tracks
Specifies the number of tracks per macro pin. This option setting is used to calculate the keepout margin based on wire tracks. When you specify -tracks_per_macro_pin, the outer keepout margin is calculated by multiplying the number of tracks by the number of pins. For example, if you set number_of_tracks to 0.5, and the macro contains 100 pins on the side, the keepout margin for that side is 50 tracks.

$100 \text{ pins} * 0.5 \text{ tracks/pin} = 50 \text{ tracks}$

The -tracks_per_macro_pin is mutually exclusive with the -inner and -outer options.

-min_padding_per_macro padding_size
Specifies the minimum padding distance per macro. This value is used to calculate the keepout margin based on pin count, and is valid only with the -tracks_per_macro_pin option. By default, the minimum padding is the minimum grid size, which is 0.

-max_padding_per_macro padding_size
Specifies the maximum padding size per macro. This value is used to calculate the keepout margin based on pin count, and is valid only with the -tracks_per_macro_pin option. By default, there is no maximum padding value.

design_and_cell_list
Specifies the list of physical designs and cells for which to create keepout margins.

DESCRIPTION

The `create_keepout_margin` command creates keepout margins for the specified physical designs and cells.

The outer and inner keepout margins can be set explicitly with the -outer and -inner options. Outer keepouts are allowed only on leaf cells and physical designs. Inner keepouts are allowed only on physical designs and are not supported on leaf cells.

Outer keepout margins can be created for macro cells based on the pin count by using the -tracks_per_macro_pin option. The minimum and maximum margins can be specified using the -min_padding_per_macro and -max_padding_per_macro options.

Multicorner-Multimode Support

This command has no dependency on scenario-specific information.

EXAMPLES

The following example creates the soft outer keepout margin on a leaf

cell.

```
prompt> create_keepout_margin -type soft \  
-outer {10 10 10 10} [get_cells MY_LEAF_CELL]
```

The following example creates the hard outer and inner keepout margins on a design.

```
prompt> create_keepout_margin -type hard \  
-outer {10 10 10 10} -inner {5 5 5 5} [get_designs MY_DESIGN]
```

The following example creates the hard outer and inner keepout margins on a physical design.

```
prompt> create_keepout_margin -type hard \  
-outer {10 10 10 10} -inner {5 5 5 5} [get_cells MY_BLOCK]
```

The following example creates the hard outer keepout margin based on the pin count of the macro.

```
prompt> create_keepout_margin -tracks_per_macro_pin .6 \  
-min_padding_per_macro .1 -max_padding_per_macro 0.2 [get_cells MY_MACRO]
```

The following example creates the routing_blockage keepout margin for the PG router on a macro cell for the specified layers.

```
prompt> create_keepout_margin type routing_blockage \  
-outer {10 10 10 10} -layers {M1 M3 M5} [get_cells MY_MACRO]
```

SEE ALSO

[get_attribute\(2\)](#)
[get_keepout_margins\(2\)](#)
[remove_keepout_margins\(2\)](#)
[report_keepout_margins\(2\)](#)

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